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High-performance p-Cu_2O/*n*-TaON heterojunction nanorod photoanodes passivated with an ultrathin carbon sheath for photoelectrochemical water splitting†

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Considerable efforts have been made to design and discover photoactive nanostructured (oxy)nitride materials that can be used as photoanodes for photoelectrochemical (PEC) water splitting. However, the high recombination rate of photoexcited electron–hole pairs and the poor photostability have greatly limited their practical applications. Herein, a p-type Cu_2O/n-type TaON heterojunction nanorod array passivated with an ultrathin carbon sheath (carbon–Cu_2O/TaON) as a surface protection layer was produced *via* a solution-based process. Due to the shape anisotropy and p–n heterojunction structure, the photocurrent density of carbon–Cu_2O/TaON heterojunction nanorod arrays as the integrated photoanode, with a maximum IPCE of 59% at a wavelength of 400 nm, reached 3.06 mA cm^{-2} under AM 1.5G simulated sunlight at 1.0 V *vs.* RHE and remained at about 87.3% of the initial activity after 60 min irradiation. Not only is the onset potential negatively shifted but the photocurrent density and photostability are also significantly improved for this photoanode compared to those of TaON and Cu_2O/TaON. These improvements are due to a high built-in potential in the p–n heterojunction device that is protected from the electrolyte by being encapsulated in an ultrathin graphitic carbon sheath. Our design introduces material components to provide a dedicated charge-transport pathway, alleviating the reliance on the materials' intrinsic properties, and therefore has the potential to greatly broaden where and how various existing materials can be used in energy-related applications.

Broader context

With the increasing interest in utilizing solar energy to diminish energy crises, solar energy-driven water splitting to form hydrogen and oxygen is one of the promising approaches that can mimic natural photosynthesis to directly harvest and convert solar energy into clean and sustainable hydrogen energy. Tantalum oxynitride (TaON) semiconductors are potential candidates for photoanode materials because of the narrow bandgap allowing visible light absorption and the appropriate band levels for water splitting. However, the high recombination rate of photoexcited electron–hole pairs and the poor photostability have greatly limited their practical applications. In the present work, we have designed and fabricated the p-type Cu_2O/n-type TaON heterojunction nanorod array passivated with an ultrathin carbon sheath as a surface protection layer. Not only is the onset potential negatively shifted but the photocurrent and photostability are also significantly improved for this product in comparison to those for TaON and Cu_2O/TaON, due to a high built-in potential in the p–n heterojunction device that is protected from the electrolyte by being encapsulated in the ultrathin graphitic carbon sheath. To help guide continuing research in this field, we have identified key challenges that must be overcome to drive down the costs of possible application of oxynitride photoanodes for PEC water splitting.

1. Introduction

Photoelectrochemical (PEC) water splitting offers the capability of harvesting the energy in solar radiation and transferring it directly to chemical bonds for easy storage, transport, and use in the form of hydrogen.[1–3] Among the various efforts involved in improving a PEC system, the appropriate choice of photoelectrode materials using solar energy is especially important because their properties, such as optical absorption characteristics and chemical stability, determine the system's performance.[1,2] Many metal oxides, such as Fe_2O_3,[4,5] TiO_2,[6,7] WO_3 (ref. 8 and 9) and $BiVO_4$,[10–12] have received immense attention as photoelectrodes for PEC conversion of solar energy into chemical fuels. However, compared to the potential of the oxygen 2p orbital in metal oxides, the more negative potential of the nitrogen 2p orbital can result in different (oxy)nitrides with

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narrow band gaps that could make use of nearly the entire solar spectrum.[13]

The (oxy)nitrides containing Ta^{5+} or Ti^{4+},[14–16] such as TaON, Ta_3N_5 and $LaTiO_2N$, have emerged as promising candidates for PEC water splitting.[17–19] TaON, as an n-type semiconductor, is a promising material that uses a large portion of the solar spectrum even without external bias and meets many of the requirements for water splitting: (i) it has a suitable band gap; (ii) it is feasible for water splitting due to appropriate band positions; and (iii) it is composed of nontoxic elements, making it environmentally benign.[20–23] Despite such promising attributes, the overall water splitting efficiency of TaON photoanodes falls well short of the theoretical maximum efficiency, which may be due to (i) its low carrier mobility, and (ii) its poor stability caused by its being subject to self-photocorrosion in electrolyte solution.[20–23]

Generally, photocatalyst features such as high crystallinity, low defect density, short charge-transfer distance, large surface area, and special morphology usually enhance the photocatalytic performance. Nanostructured photoelectrodes have begun to address limitations of current materials. One-dimensional (1D) nanostructured arrays have the apparent advantages of promoting the transport and separation of photoexcited charge carriers, and providing abundant surface reaction sites, which are crucial for obtaining high solar energy conversion efficiency.[24–27] For example, polycrystalline Ta_3N_5 nanotube arrays on a Ta substrate were prepared by anodization of tantalum foil.[28–30] Ta_3N_5 nanorod arrays were produced *via* through-mask anodization and subsequent nitridation using anodic alumina as the mask under the anodizing voltage of 650 V, a vapor-phase hydrothermal process and subsequent nitridation.[31,38] Besides Ta_3N_5 photoanodes, 1D TaON nanotube arrays can also be prepared,[32] but it is still challenging to synthesize a desired 1D TaON nanostructured array using low-cost and easily-scalable routes.

For efficient water splitting using photoanodes, one prospective solution is use of a p–n photochemical diode that promotes the separation of photogenerated electrons and holes, because a single material is unlikely to possess both the narrow bandgap and proper band positions required. A junction is a type of design that can be used to separate electrons and holes by an internal electric field induced from band bending. For example, the heterojunction films, such as various WO_3/$BiVO_4$,[33] $CaFe_2O_4$/$ZnFe_2O_4$,[34] Fe_2O_3/$ZnFe_2O_4$,[35] and $CaFe_2O_4$/TaON,[36] have shown promising photoelectrochemical activity, with the improvement ascribed to efficient carrier separation in the heterojunction. In particular, cuprous oxide, Cu_2O, is an attractive p-type oxide for photoelectrochemical hydrogen production with a direct bandgap of 2.0 eV and a corresponding theoretical photocurrent density of 14.7 mA cm^{-2} and a light-to-hydrogen conversion efficiency of 18% based on the AM1.5 spectrum. These attractive features are due to Cu_2O possessing favourable energy band positions, with the conduction band lying 0.7 V negative of the hydrogen evolution potential and the valence band lying just positive of the oxygen evolution potential.[37] Since TaON has the appropriate conduction and valence band edges, the band gap of TaON is well-matched with Cu_2O for efficient charge carrier separation. Therefore, developing Cu_2O/TaON arrays as a p–n heterojunction photoanode for PEC water splitting is highly desirable.

The practical application of (oxy)nitride photoelectrodes is seriously hindered by poor photostability. To solve this problem, most efforts with regards to Ta_3N_5 photoanodes have involved alleviating accumulation of photogenerated holes by facilitating water oxidation using various oxygen evolution reaction (OEC) cocatalysts.[38–42] However, it is also essential to provide highly stable TaON-based photoelectrodes coupled with a protective layer.[18,19] Besides the OEC catalysts, carbon is a fascinating material that has offered exciting new opportunities in various fields of nanotechnology including photocatalysis based on properties such as high electrical conductivity, non-toxicity, high strength, and high chemical stability.[43,44] In particular, it is easy to coat the carbon layer on an unstable semiconductor-based photoelectrode in a solution-based carbon precursor.[43–46] For example, Zhu *et al.* reported that surface hybridization of TiO_2 with graphite-like carbon layers of a few molecular layers thickness yields efficient photocatalysts.[43] Alper *et al.* reported about using an ultrathin graphitic carbon layer on porous silicon nanowires with high specific capacitance to produce a planar microsupercapacitor electrode.[45] To the best of our knowledge, however, no previous work has been reported regarding the application of ultrathin carbon sheath-passivated Cu_2O/TaON photoelectrodes with perfect photostability for efficient photoelectrochemical water oxidation.

The requirement for nanostructured TaON with the low recombination of electron–hole pairs and good photostability, presents a potential challenge for coupling many well-known water splitting catalysts. In this work, we report on the fabrication of aligned TaON nanorod arrays using a general vapor-phase hydrothermal route with subsequent nitridation. After the involvement of a $Cu(OH)_2$/TaON nanorod array prepared by a chemical bath deposition process into a carbon precursor-based solution, the Cu_2O nanoparticle/TaON nanorod array passivated with an ultrathin carbon sheath as a surface protection layer against photocorrosion (carbon–Cu_2O/TaON) was successfully produced *via* a subsequent heat-treatment strategy. This production of an integrated photoanode consisting of carbon–Cu_2O/TaON nanorod arrays protected with a metal/co-catalyst-free carbon layer is highly cost-effective and easily scalable. Moreover, solar water splitting using this photoanode achieved superior enhancement of the photoelectrochemical (PEC) performance, including the best durability against photocorrosion to date. The durability of this photoanode is one of the highest among all the photoanodes so far reported in the field of energy conversion.

2. Experimental section

2.1 Methods

2.1.1 Fabrication of TaON nanorod arrays. A Ta foil with a thickness of 0.25 mm (Alfa Aesar) was washed in ethanol, acetone, isopropanol and deionized water each for 60 minutes before use. The clean Ta foil was suspended above a 0.15 M HF aqueous solution in a Teflon-lined autoclave, which was then heated at 180–240 °C for 0–6 h to grow F-containing Ta_2O_5 (F–Ta_2O_5) nanorod arrays on the Ta foil.[40] After this heat treatment, the resultant F–Ta_2O_5 nanorod arrays on the Ta foil were placed under a gaseous atmosphere of NH_3 with a flow of 20 mL min^{-1} and heated at 650–750 °C for 3 h, producing the TaON nanorod arrays.

2.1.2 Fabrication of carbon–Cu_2O/TaON nanorod arrays. A chemical bath deposition process was developed to decorate the surfaces of the TaON nanorod arrays with Cu_2O nanocrystals. Typically, as-prepared TaON nanorod arrays were added into an ethanolic solution of 0.04 M $Cu(CH_3COO)_2$, and the resulting adsorption and deposition of Cu^{2+} ions onto the surfaces of the TaON nanorod arrays were allowed to take place for different time intervals (10, 20, 40 and 60 s). After the Cu^{2+} ion adsorption, the TaON nanorod arrays were put into a NaOH solution, and maintained at 60 °C in a water bath for the formation of the $Cu(OH)_2$/TaON nanorod arrays. Then, an aqueous solution of glucose (5 mL) as a carbon precursor at various concentrations (0.01, 0.05, 0.1 and 0.2 M) was slowly added into the above suspension. After ultrasonication for a certain amount of time, the samples were rinsed with distilled water to remove CH_3COO^-, glucose, and NaOH, and then dried under vacuum at a temperature of 60 °C for 2 h. Finally, the $Cu(OH)_2$/TaON nanorod arrays with or without glucose were immediately transferred into the furnace and annealed at 500 °C in N_2 atmosphere for two hours, leading to the production of the carbon–Cu_2O/TaON and Cu_2O/TaON nanorod arrays.

2.2 Characterization

The obtained products were characterized with powder X-Ray diffraction (XRD, MAC Science Co. Ltd Japan) using Cu K_α (λ = 0.1546 nm) for the XRD patterns. The morphology and size of the resultant powders were characterized by a Zeiss Ultra 55 field-emission scanning electron microscope (SEM) associated with an X-Ray energy-dispersive spectrometer (EDX). Transmission electron microscopy (TEM) images were visualized using a transmission electron microscope (TEM, JEM-2010) at an acceleration voltage of 200 kV. The chemical states of the sample were determined by X-Ray photoelectron spectroscopy (XPS) in a VG Multilab 2009 system (UK) with a monochromatic Al K_α source and charge neutralizer. The optical properties of the samples were analyzed by UV-Vis diffuse reflectance spectroscopy using a UV-Vis spectrophotometer (UV-2550, Shimadzu). Raman spectroscopy was conducted with a Renishaw LabRAM confocal Raman microscope utilizing a 633 nm wavelength laser.

2.3 Photoelectrochemical water splitting

The photoelectrochemical water splitting was carried out in a three-electrode system, where the TaON, Cu_2O/TaON and carbon–Cu_2O/TaON photoanodes (the irradiation area was 1 cm^2), a saturated calomel electrode (SCE) and a high-surface area platinum mesh act as the working electrode, reference electrode and counter electrode, respectively. An aqueous solution of 0.5 M NaOH was used as the electrolyte (pH = 13.6, 100 mL). The electrolyte was stirred and purged with Ar gas before the measurements. The photoanodes were illuminated with AM 1.5G simulated sunlight (100 mW cm^{-2}) from a commercial solar simulator. The measured potential *versus* SCE was converted to the RHE scale according to the Nernst equation ($E_{RHE} = E_{SCE}$ + 0.059 pH + 0.242), where E_{RHE} is the potential *vs.* a reversible hydrogen potential, E_{SCE} is the potential *vs.* the SCE electrode, and pH is the pH value of electrolyte. The wavelength-dependent incident photon-to-current efficiency (IPCE) was calculated according to the following equation:

$$\mathrm{IPCE}(\%) = \frac{1240 \times i_{\mathrm{ph}}}{\lambda \times p_{\mathrm{in}}} \times 100$$

where i_{ph} is the photocurrent (mA), λ is the wavelength (nm) of incident radiation, and p_{in} is intensity of the incident light irradiating on the semiconductor electrode at the selected wavelength (mW).[47] Photocurrent–potential curves were measured at a rate of 30 mV s^{-1}. The wavelength dependence of the IPCE was measured under monochromatic irradiation from a Xe lamp equipped with band-pass filters (central wavelengths: 400, 420, 440, 460, 480, 500, 540, 580 and 600 nm; FWHM: 10 nm).

The evolution of hydrogen and oxygen by photoelectrochemical water splitting was conducted in an air-tight reactor connected to a closed gas circulation system. The TaON, Cu_2O/TaON and carbon–Cu_2O/TaON photoanodes were biased at 1.0 V *vs.* RHE in a stirred aqueous solution of 0.5 M NaOH (pH = 13.6) under AM 1.5G simulated sunlight. The amount of hydrogen or oxygen was determined by gas chromatography (GC-3240, TCD, Ar carrier). For Faradaic efficiency, a two-electrode cell (no reference electrode) was used to measure the Faradaic efficiency. A carbon–Cu_2O/TaON photoanode and a Pt foil were used as a working electrode and a counter electrode, respectively. The bias was 1.0 V. The area of the carbon–Cu_2O/TaON was about 1 cm^2. The cell was sealed and was purged by Ar for half an hour and no O_2 or N_2 was detected before the Faradaic efficiency measurement.

3. Results and discussion

The synthesis of the integrated photoanode made of Cu_2O nanoparticles/TaON nanorod array passivated with an ultrathin carbon sheath as a surface protection layer is presented in Scheme 1. Ta foil was selected as the base material for the photoanode because it is a conductive substrate and a direct precursor of the F–Ta_2O_5 nanorod array grown on the Ta substrate *via* a vapour-phase hydrothermally induced self-assembly technique (Fig. S1–S2†). The TaON nanorod array with its unique open area was then obtained by the subsequent nitridation process. By employing a chemical bath deposition process, the $Cu(OH)_2$ species was formed on the surface of the TaON nanorod array by dissolution of Cu^{2+} ions into a NaOH solution. To obtain the desired 1D nanorod structure, it is imperative to maintain the Cu_2O/TaON nanorod array after the deposition of Cu_2O nanoparticles. Finally, the delicate conversion of $Cu(OH)_2$ to Cu_2O nanoparticles on the surface of the

TaON nanorod array along with the simultaneous formation of a thin carbon sheath as a protective layer, as shown in Scheme 1, was achieved by a solution-based carbon precursor coating of the initial $Cu(OH)_2$/TaON nanorod array and subsequent annealing of the glucose–$Cu(OH)_2$/TaON nanorod array under inert N_2 environment, resulting in the formation of the carbon–Cu_2O/TaON nanorod array as an integrated photoanode.

The top surfaces of TaON, Cu_2O/TaON, and the carbon–Cu_2O/TaON nanorods array were viewed using field-emission scanning electron microscopy (FESEM). As shown in Fig. 1a, the TaON nanorods array were observed to be vertically aligned on the entire surface of the Ta substrate with uniform nanorods that have an average diameter of 40–60 nm. This observation indicates that the surface of the individual TaON nanorod array is smooth, as shown previously.[39,40] After the chemical bath deposition process and the subsequent heat-treatment of the $Cu(OH)_2$/TaON nanorod array, the Cu_2O nanoparticles with an ultra-small particle size were homogeneously formed on the surface of the TaON nanorod array, as shown in Fig. 1b. During the phase-transformation process, the conversion of $Cu(OH)_2$ to Cu_2O is presumably achieved by dehydration of $Cu(OH)_2$ to CuO, followed by removal of oxygen from the lattice of CuO to form Cu_2O at high temperatures in inert N_2 atmosphere.[44] Furthermore, the $Cu(OH)_2$/TaON nanorod array was added into the glucose solution that was used as a carbon precursor, and the carbon sheath-covered Cu_2O/TaON nanorod array was produced *via* the subsequent heat-treatment strategy, as shown in Fig. 1c. At the initial stage of the heat-treatment process, the glucose dehydrates and cross-links, and as the reaction continues, aromatization and carbonization take place; the resulting firm carbonized shell covering the nanorod surfaces maintains the integrity of the morphology of the initial $Cu(OH)_2$ species.[44,48] Note that upon close observation of the composites of carbon–Cu_2O/TaON nanorod arrays, the Cu_2O nanoparticles are still clearly visible after the carbon coating, indicating that the carbon layer is ultrathin. In order to understand the structure of this heterojunction, a schematic illustration of the carbon–Cu_2O/TaON nanorod array is shown (see Fig. 1d). To confirm the quality of the entire carbon–Cu_2O/TaON nanorod array, a highly magnified SEM image of the surface is also presented (see Fig. S3†); this image indicates the entire carbon–Cu_2O/TaON nanorod array retained the homogeneous 1D morphology after the modification of the Cu_2O nanoparticles and the ultrathin carbon layer. Thus, the rationally designed and synthesized carbon–Cu_2O/TaON nanorod array is expected to work as an efficient and stable photoanode for water splitting applications.

The obtained TaON, Cu_2O/TaON and carbon–Cu_2O/TaON nanorod arrays were further characterized using transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM). TEM confirms the single-crystal nature of the TaON nanorods and their preferential growth of the [001] direction (Fig. 2). The TEM images of the Cu_2O/TaON heterojunction indicate that the decoration of the Cu_2O

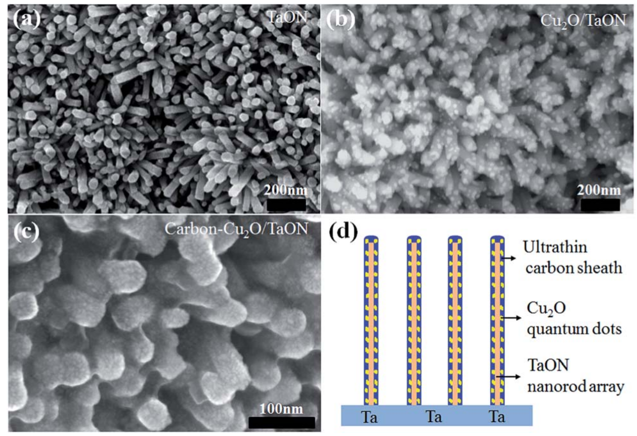
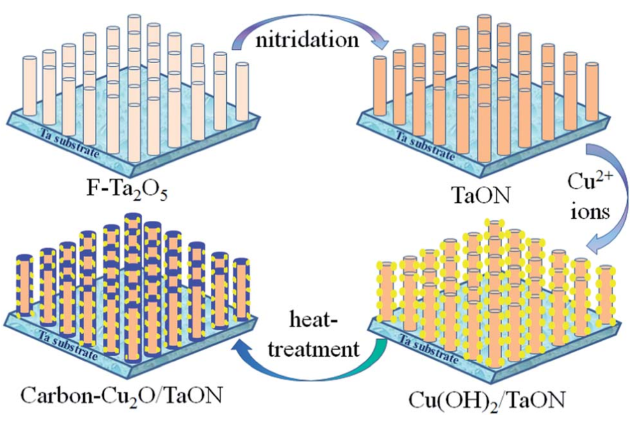


Fig. 1 FESEM images of the top surface of (a) TaON, (b) Cu_2O/TaON, and (c) a carbon–Cu_2O/TaON nanorod array. (d) Schematic illustration of an array of carbon–Cu_2O/TaON nanorods.



Scheme 1 Schematic illustration of ultrathin carbon sheath-passivated Cu_2O/TaON nanorods array.

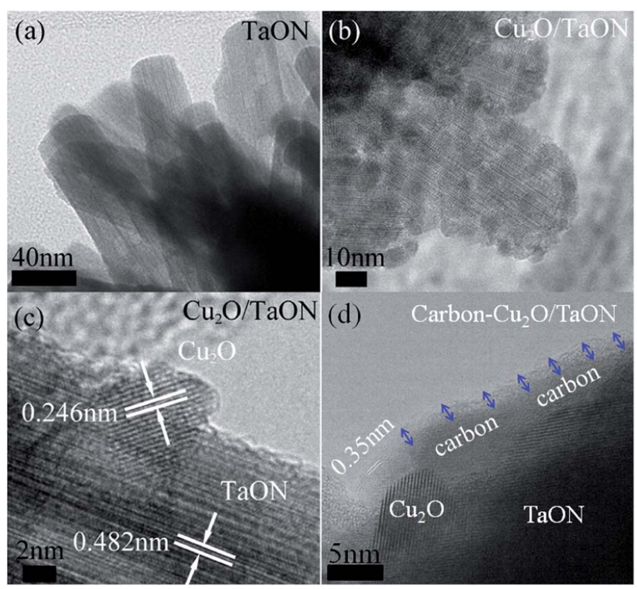


Fig. 2 TEM (a and b) and HRTEM (c and d) images of (a) TaON, (b and c) Cu_2O/TaON, and (d) carbon–Cu_2O/TaON.

nanoparticles yielded average particle sizes ranging from 3 to 10 nm, and that these decorated nanoparticles were homogeneously dispersed on the surface of the TaON nanorods. The HRTEM images of Cu_2O/TaON (Fig. 2b and c) show that these interactions lead to the formation of an intact interface between Cu_2O and TaON, where the 0.246 nm and 0.482 nm lattice fringes corresponded to the reflections from the (111) plane of Cu_2O (JCPDS 05-667) and the reflections from the (100) plane of TaON (JCPDS 70-1193). Fig. 2d shows a high-resolution (HR) TEM image of carbon–Cu_2O/TaON, which was used to estimate the thickness of the carbon shell, and shows that the Cu_2O/TaON nanorod array is surrounded by a few graphitic carbon layers, and the overall thickness of the carbon shell is about ~3 nm. The *d*-spacing of the graphitic layers is 0.35 nm, which is in agreement with the literature. The scanning transmission electron microscope (STEM)–energy dispersive X-Ray spectroscopy (EDX) elemental mapping (Fig. 3) shows the carbon–Cu_2O/TaON nanorods with an average diameter of 40–60 nm. In contrast, as also seen in Fig. 3, the carbon sheath and Cu_2O nanoparticles in the carbon–Cu_2O/TaON configuration were observed to be very uniform, with the brighter spots in the Ta, Cu, C, O and N elemental mapping showing the distribution of Cu_2O and carbon species. Thus, since the surface of the Cu_2O/TaON nanorod array is fully covered with a thin graphite-like carbon film, a good electronic contact between (oxy)nitride semiconductor and carbon was achieved.

The phase purity, crystal structure, chemical states of each element and optical properties of the obtained samples were examined by X-Ray diffraction (XRD), X-Ray photoelectron spectroscopy (XPS) and UV-Vis diffuse reflectance spectroscopy (UV-Vis DRS). After the chemical bath deposition process, the $Cu(OH)_2$/TaON nanorod array was obtained. According to the XRD patterns in Fig. 4S,† five main diffraction peaks near or at $2\theta = 23.8$, 34.1, 35.9, 39.8 and 53.2° can be observed, corresponding to the (021), (002), (111), (130) and (132) planes of $Cu(OH)_2$ [JCPDS no. 12-420, orthorhombic, space group: *Cmc*2$_1$], respectively, and confirming the presence of the $Cu(OH)_2$ phase on the surface of the TaON nanorod array (Fig. S4†). After the subsequent heat-treatment of the $Cu(OH)_2$/TaON nanorod array, three main diffraction peaks near or at $2\theta = 36.4$, 42.5 and 61.3° can be observed, corresponding to (111), (200) and (220) planes of Cu_2O [JCPDS no. 05-667, cubic, space group: *Pn*3*m*], respectively, and confirming the transformation of the $Cu(OH)_2$ to the Cu_2O phase on the surface of the TaON nanorod (Fig. S4†). Moreover, there are no obvious differences in the intensities and widths of the XRD peaks from Cu_2O/TaON and from carbon–Cu_2O/TaON, due to the low loading content of carbon. Note especially that the characteristic diffraction peaks of the TaON/Ta phase (JCPDS no. 70-1193) are observed and there are no obvious changes of intensities and widths in all samples. In addition, the two strong peaks from the (211) and (321) planes can be indexed to the Ta phase for all samples. This implies that there is no significant change observed in phase structure and crystallite size of TaON after the co-modification of Cu_2O and carbon. The XRD patterns of the carbon–Cu_2O/TaON sample indicate that the deposited Cu_2O and carbon species are attached to the surface of the TaON nanorod array, which is in agreement with the SEM and TEM measurements.

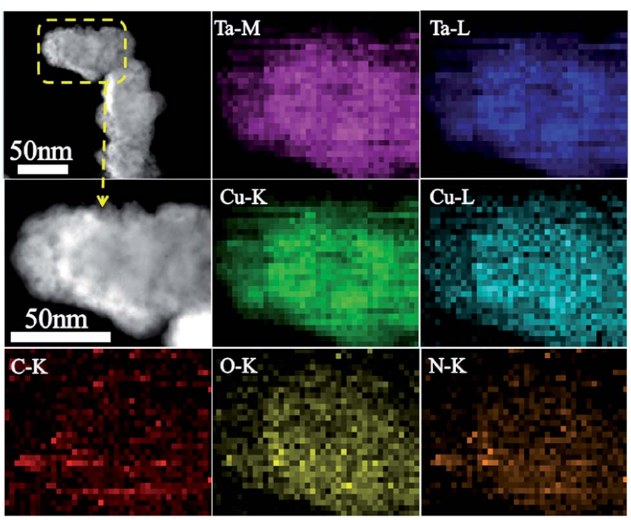


Fig. 3 STEM images and corresponding EDX elemental mapping of Ta, Cu, C, O and N of the carbon Cu_2O/TaON nanorods.

In order to determine the surface composition and chemical nature of the carbon–Cu_2O/TaON composites, the chemical state of each element in the samples was carefully checked by X-Ray photoelectron spectroscopy (XPS). The XPS survey spectrum revealed the presence of Ta, O, N, C and Cu elements (Fig. 4 and S5 and S6†). The binding energies were calibrated by using the contaminating carbon C1s peak at 284.5 eV as a standard. XPS signals of Ta 4f in the carbon–Cu_2O/TaON composites (see Fig. S5†), observed at binding energies at around 26.0 eV (Ta $4f_{7/2}$) and 27.6 eV (Ta $4f_{5/2}$), are ascribed to Ta^{5+}.[20–23] Fig. 4b displays the Cu 2p level spectrum. The Cu $2p_{3/2}$ and Cu $2p_{1/2}$ spin-orbital photoelectrons were observed to be located at binding energies of 932.5 eV and 952.4 eV, respectively, which are in good agreement with the reported values of Cu_2O.[44] Obviously, the deposited nanoparticles were Cu_2O rather than Cu or CuO. Especially, after the addition of the ultrathin carbon onto the surface of Cu_2O/TaON, the strong C–C peak at 285.0 eV implies the formation of a carbon layer on the surface of Cu_2O/TaON, as shown in Fig. 4c. Furthermore, a notable decrease in oxygen content is clearly visible and the peak corresponding to the C–O bond has disappeared (see Fig. 4c), compared with the extensive results of carbon.[49] The oxygen loss mainly results from the loss of C–O and O–C═O, indicating the partial removal of the oxygen-containing functional groups, which is in agreement with the literature.[49]

For the optical properties of the samples, it was observed that the absorption onset of the pure TaON nanorod array was at a wavelength of approximately 500–550 nm, which agreed well with the bandgap of the TaON sample ($E_g = 2.4$ eV).[14] With regard to Cu_2O/TaON, due to the narrow bandgap of Cu_2O ($E_g = 2.0$ eV),[37] the Cu_2O/TaON nanorod array displayed red shifts in the bandgap transition and an enhanced absorption in the visible-light region, shown in Fig. 4d. After the coating of the ultrathin carbon sheath using the solution-based carbon precursor coating (*i.e.*, glucose solution) combined with

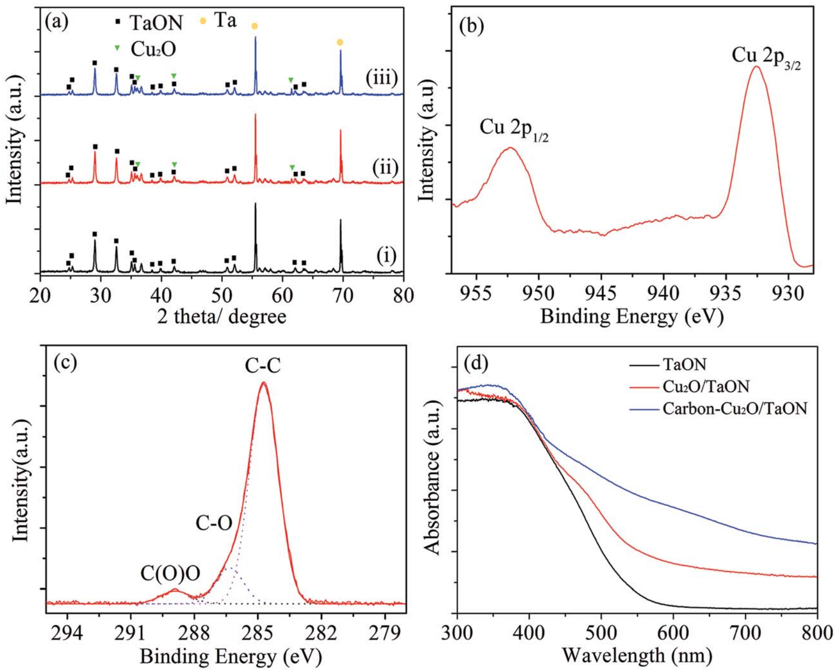


Fig. 4 (a) XRD patterns of (i) TaON, (ii) Cu_2O/TaON and (iii) carbon–Cu_2O/TaON photoanodes, (b and c) XPS spectra of a carbon–Cu_2O/TaON photoanode and (d) UV-Vis diffuse reflectance spectra of TaON, Cu_2O/TaON and carbon–Cu_2O/TaON photoanodes.

subsequent carbonization, the carbon–Cu_2O/TaON showed enhanced absorption in the long-wavelength region due to the strong absorption of the carbon layer. This suggests an increased electric surface charge of the semiconductor within the carbon–Cu_2O/TaON composite due to the introduction of the carbon, which can possibly cause modifications of the fundamental process of electron–hole pair formation during irradiation.

PEC measurements were carried out in a three-electrode system, with the TaON nanorod arrays as the working electrode with an exposed area of 1 cm^2, a Pt foil as the counter electrode, a saturated calomel electrode (SCE) as a reference electrode, and the NaOH electrolyte (0.5 M, pH = 13.6). The photocurrent density *versus* applied voltage scans of the samples were measured under AM 1.5G simulated sunlight (100 mW cm^{-2}). The bare TaON photoanode showed negligible current in the dark (see Fig. 5a). However, the current density of the bare TaON nanorod array photoanode reached 0.63 mA cm^{-2} at 1.0 V *vs.* RHE and even 0.93 mA cm^{-2} at 1.23 V *vs.* RHE under AM 1.5G simulated sunlight (100 mW cm^{-2}). This low current density is ascribed to the high levels of recombination and serious photocorrosion during the slow interface charge transfer since the photogenerated holes are accumulated on the surface of the TaON photoanode. After the modification of the Cu_2O nanoparticles, the Cu_2O/TaON nanorod array yielded a photocurrent density of 2.12 mA cm^{-2} at 1.0 V *vs.* RHE and even 2.94 mA cm^{-2} at 1.23 V *vs.* V_{RHE} under AM 1.5G solar light, which are higher than that of the $CaFe_2O_4$/TaON photoanode (1.26 mA cm^{-2} at 1.23 V *vs.* V_{RHE}).[17] Moreover, the Cu_2O/TaON nanorod array passivated with an ultrathin carbon sheath (carbon–Cu_2O/TaON) yielded a photocurrent density of 3.06 mA cm^{-2} at 1.0 V *vs.* RHE and even 4.36 mA cm^{-2} at 1.23 V *vs.* V_{RHE} under AM 1.5G solar light, which are almost five times higher than that of the bare TaON photoanode. It is evident that the sharp rise beyond the onset potential and the increased photocurrent density in the high-potential region indicate high charge transfer and collection efficiency for the carbon–Cu_2O/TaON photoanode. Moreover, the photocurrent density of the carbon–Cu_2O/TaON heterostructured photoanode is 1.5–5.4 mA cm^{-2} in the potential range of 0.8–1.5 V *vs.* RHE under AM 1.5G 100 mW cm^{-2} simulated sunlight, exhibiting that not only the onset potential is negatively shifted but also that the photocurrent is significantly improved due to the effective charge-transfer process.

To verify the above photocurrents under AM 1.5G simulated sunlight condition, the incident photon-to-current efficiencies (IPCE) of the TaON, Cu_2O/TaON and carbon–Cu_2O/TaON photoanodes have been measured under monochromatic light irradiation and plotted as a function of wavelength at a given voltage (Fig. 5b). At an applied potential of 1.0 V *vs.* RHE, the IPCEs for the carbon–Cu_2O/TaON photoelectrode were measured to be above 48% for the incident light wavelength range of 400–440 nm, which is higher than that of CoO_x/TaON electrode (*ca.* 42% at a wavelength of 400 nm at 1.2 V *vs.* RHE in sodium phosphate buffer solution) and that of $CaFe_2O_4$/TaON electrode (*ca.* 30% at a wavelength of 400 nm at 1.23 V *vs.* RHE in 0.5 M NaOH solution) in the previous work.[17,18] However, it is lower than that of IrO_x/TaON electrode (*ca.* 76% at a wavelength of 400 nm at 1.15 V *vs.* RHE in Na_2SO_4 solution), while the photocurrent of the IrO_x/TaON photoanode was decreased markedly due to the self-oxidative deactivation of the TaON surface.[19] In addition, in the wavelength range of 500–600 nm,

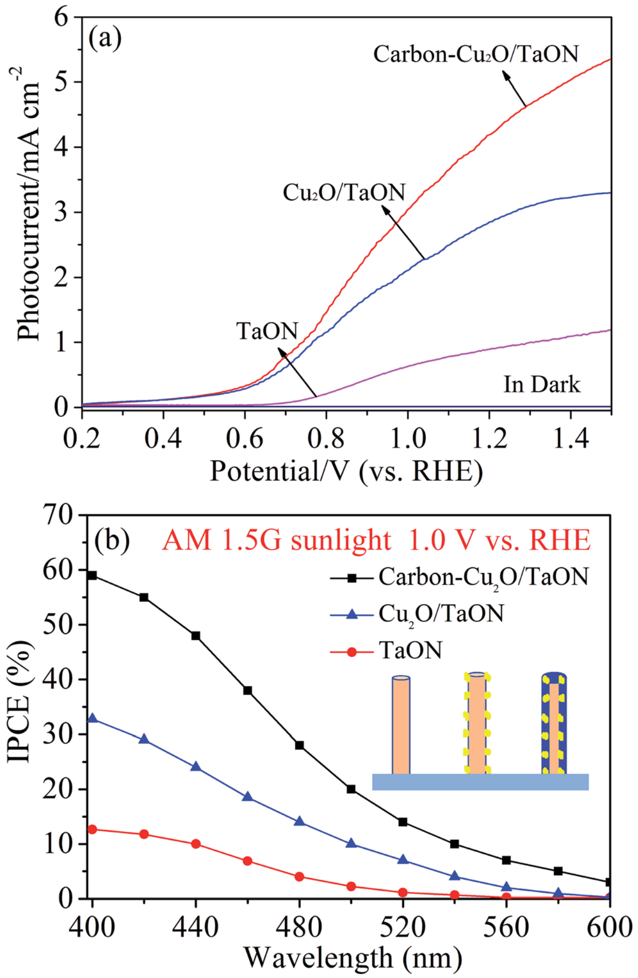


Fig. 5 (a) Photocurrent–potential curves in the dark and under AM 1.5G simulated sunlight (100 mW cm^{-2}), and (b) the incident photon-to-current efficiency (IPCE) at 1.0 V *vs.* RHE under monochromatic irradiation for TaON, Cu_2O/TaON and carbon–Cu_2O/TaON photoanodes.

the IPCEs for the carbon–Cu_2O/TaON photoelectrode were measured to be 3–20% while they were below 2.2% for the TaON photoanode. These measurements indicate that the carbon–Cu_2O/TaON photoanode displayed enhanced IPCEs in the long-wavelength region due to the red shifts in the bandgap transition from the introduction of Cu_2O nanoparticles and due to the strong absorption of the carbon sheath. This enhancement of the IPCEs is in agreement with the distribution of the UV-Vis diffuse reflectance spectra. It is worth pointing out that a maximum IPCE of 59% was achieved at a wavelength of 400 nm for the carbon–Cu_2O/TaON photoanode. Thus, the combined results demonstrate that the recombination of photoexcited carriers is effectively inhibited through the modification of the Cu_2O nanoparticles and ultrathin carbon layer; these modifications can accelerate the water splitting reaction and improve the charge transfer processes.

The practical application of (oxy)nitride photoelectrodes during the water splitting reaction is hindered due to the poor photocurrent stability.[13–19] A major aim of this work, in addition to enhancing the photocurrent density through the TaON nanorod array, is to improve the photostability of the TaON nanorod array photoanode. The effect of Cu_2O nanoparticles and ultrathin carbon layer on the photostability of TaON nanorod array was examined by tracking the photocurrent densities of bare TaON, Cu_2O/TaON and carbon–Cu_2O/TaON photoelectrodes in NaOH (pH = 13.6) solution at 1.0 V *vs.* RHE under AM1.5G illumination (100 mW cm^{-2}) for 60 min. As shown in Fig. 6a, the photocurrent generated by the bare TaON array decreased significantly within a short period of time, and the steady state photocurrent became negligibly low within a few seconds, indicating that photogenerated holes accumulated at the surface of TaON due to poor kinetics for water splitting and subsequent surface oxidative decomposition of TaON.[20–23] The Cu_2O/TaON photoanode exhibited almost 46.7% of the initial photocurrent density of ~2.12 mA cm^{-2}, indicating that modification of the TaON electrode with Cu_2O nanoparticles can yield improved albeit limited photostability in an NaOH solution. To determine the cause of this limited improvement,

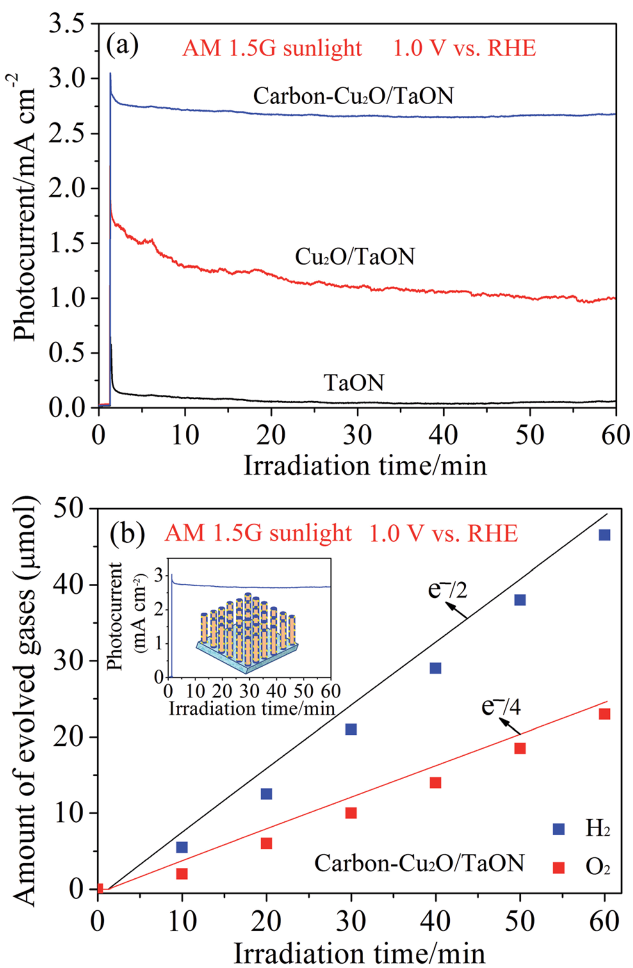


Fig. 6 Time courses for (a) the photocurrent densities of the TaON, Cu_2O/TaON and carbon–Cu_2O/TaON nanorod arrays and for (b) the amounts of oxygen and hydrogen evolved from the carbon–Cu_2O/TaON photoanodes as determined by gas chromatography. Both experiments were carried out at 1.0 V *vs.* RHE under AM 1.5G simulated sunlight at 100 mW cm^{-2}.

Auger Cu LMM spectra (Fig. S7†) were used to show the surface change after the stability test. No peak related to Cu(0) emerged in the Cu LMM spectra of Cu_2O/TaON and carbon–Cu_2O/TaON after the stability test. Combined with the previous work about poor stability of Cu_2O,[37,44] the limited enhancement of photostability for the Cu_2O/TaON photoanode compared to the bare TaON photoanode is ascribed to phase conversion, where accumulated surface holes lead to the oxidation of Cu_2O and a certain amount of photocorrosion of TaON because Cu_2O nanoparticles do not fully cover the TaON nanorods. In contrast to the limited photostabilty of the Cu_2O/TaON photoanodes, the carbon–Cu_2O/TaON photoanodes exhibited almost 87.3% of the initial photocurrent density of ~3.06 mA cm^{-2} for 60 min, indicating that the ultrathin carbon layer efficiently restrains the recombination pathways and improves the kinetic transport of photogenerated charge carriers for the carbon–Cu_2O/TaON photoanode. To understand this process further, Cu 2p XPS spectra were obtained for the carbon–Cu_2O/TaON photoanode, as shown in Fig. S8b,† after the Faradaic efficiency test. The peaks at 932.4 and 952.2 eV correspond to the binding energies of Cu $2p_{3/2}$ and Cu $2p_{1/2}$ of Cu_2O, and those at 934.3 and 953.8 eV are attributed to CuO,[51,52] confirming the coexistence of a trace amount of CuO. The small decrease of the photocurrent in the carbon–Cu_2O/TaON photoanode may thus be ascribed to the low levels of oxidation of Cu_2O and of charge recombination, as well as to low levels of other destabilizing mechanisms, such as photon exchange, tunneling through the electric potential barrier near the interface, or thermionic emission.[50] In-depth investigation of these mechanisms is beyond the scope of our present work, but efforts will be conducted in the future. Therefore, compared with that of $CaFe_2O_4$/TaON, CoO_x/TaON and IrO_x/TaON photoanodes,[17–19] this modified approach using the Cu_2O nanoparticles and ultrathin carbon layer has presented a significant enhancement of the photocurrent and stability of the TaON-based photoanodes due to a high built-in potential in the protection of the p–n heterojunction device from the electrolytes by encapsulation of the device in an ultrathin graphitic carbon sheath.

To further explore the effect of the content of Cu_2O nanoparticles and carbon on the photocurrent densities and photostabilities of the Cu_2O/TaON and carbon–Cu_2O/TaON photoanodes, these properties were measured as a function of the deposition time of Cu^{2+} and the glucose concentration, as shown in Fig. S9 and S10.† The Cu_2O/TaON and carbon–Cu_2O/TaON photoanodes using the 40 s deposition time of Cu^{2+}, and the 0.1 M glucose concentration, exhibited the best PEC performance. At these conditions, the carbon–Cu_2O/TaON photoanode demonstrated the highest photocurrent density, *i.e.*, 4.36 mA cm^{-2} at 1.23 V *vs.* V_{RHE}, and the best photostability, *i.e.*, 87.3%, at 1.0 V *vs.* V_{RHE} under AM 1.5G solar light; these values are ~4.69 times the photocurrent density and ~14.55 times the photostability than shown by the bare TaON photoanode. The Cu_2O/TaON photoelectrode exhibited lower photocurrent densities and photostabilities than did the carbon–Cu_2O/TaON photoanode at each of the various concentrations of glucose and deposition times. These results indicate that combining the Cu_2O nanoparticles and the carbon sheath, at appropriate levels, is indeed an effective strategy for improving the photocurrent density and combating photocorrosion of the TaON photoanode that we have employed. Thus, the rationally designed and synthesized carbon–Cu_2O/TaON nanorod array is expected to work as an efficient and stable photoanode for water splitting applications because (i) the TaON nanorod array with a high surface area provides a large area for contact with electrolyte; (ii) the 1D p-Cu_2O/*n*-TaON heterostructures facilitate carrier diffusion; (iii) the ultrathin carbon sheath serves as a protective film, preventing the electrolyte from contacting the semiconductors; and (iv) the carbon sheath facilitates charge transfer into the electrolyte solution during photoelectrochemical water splitting.

To verify that the measured photocurrent of the carbon–Cu_2O/TaON photoanode originates from water splitting rather than any other undesired side reactions, the oxygen evolution reaction and hydrogen evolution reaction were examined (Fig. 6b). The carbon–Cu_2O/TaON photoanode was held at 1.0 V *versus* RHE under AM 1.5G for 60 min. The amounts of H_2 and O_2 evolved from the carbon–Cu_2O/TaON heterostructured photoanode were 46.5 and 23.1 μmol cm^{-2}, respectively. The ratio of evolution rates of H_2 and O_2 is close to the stoichiometric value of 2.0. The obtained Faradaic efficiencies of 93% and 90%, determined by the measurement of the evolved H_2 and O_2 gas, respectively, indicate that the photocurrent is indeed due to the oxygen evolution reaction (OER) and hydrogen evolution reaction (HER). Thus, the above-mentioned results regarding the integrated carbon–Cu_2O/TaON photoanode indicate that the combination of the Cu_2O nanoparticles and ultrathin carbon layer can not only shorten the transport distances for the charge carriers, but also passivate some surface defects, efficiently suppressing the bulk recombination and surface recombination of the photogenerated charges.

The interfacial properties between the photoelectrodes (*i.e.*, the TaON, Cu_2O/TaON and carbon–Cu_2O/TaON nanorod arrays) and the electrolyte were scrutinized by electrochemical impedance spectroscopy (EIS) measurement that was carried out covering the frequency range of 10^4–0.1 Hz using an amplitude of 10 mV at the open circuit potential of the system. A semicircle (*i.e.*, the arch in the present study) in the Nyquist plots, where *x*- and *y*-axes are the real part (Z') and the negative of imaginary part ($-Z''$) of the impedance, represents the charge-transfer process, while the diameter of the semicircle reflects the charge-transfer resistance (Fig. 7). It is well konwn that a large value of impedance indicates a poor conductivity along the electron pathway in the photoelectrode, so it is essential to conduct the fitting of raw data to an equivalent circuit model for the EIS analysis. In this work, the obtained results were fitted into the models of two RC circuits, as shown in Fig. 7. The RC circuit with the largest resistance usually represents the electrode–electrolyte interface, and the other circuit is related to the electron transport inside the electrode. The first RC circuit presents the electron transport from TaON nanorods arrays to Ta substrate. Clearly, the arches for TaON nanorods arrays under AM 1.5G simulated sunlight were much smaller than those for TaON in the dark because the impedance of TaON-as deposited electrode was too large as compared to the other

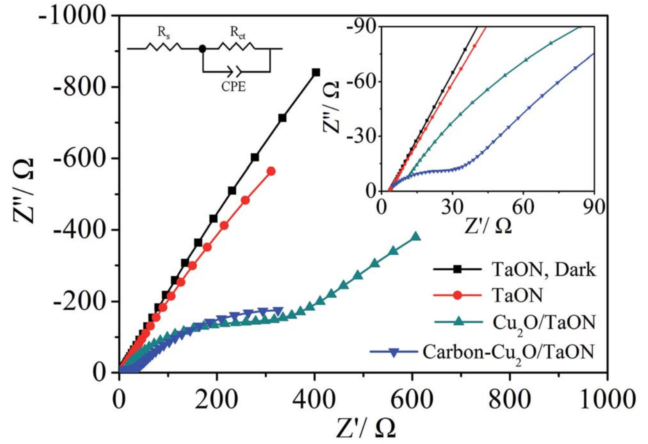
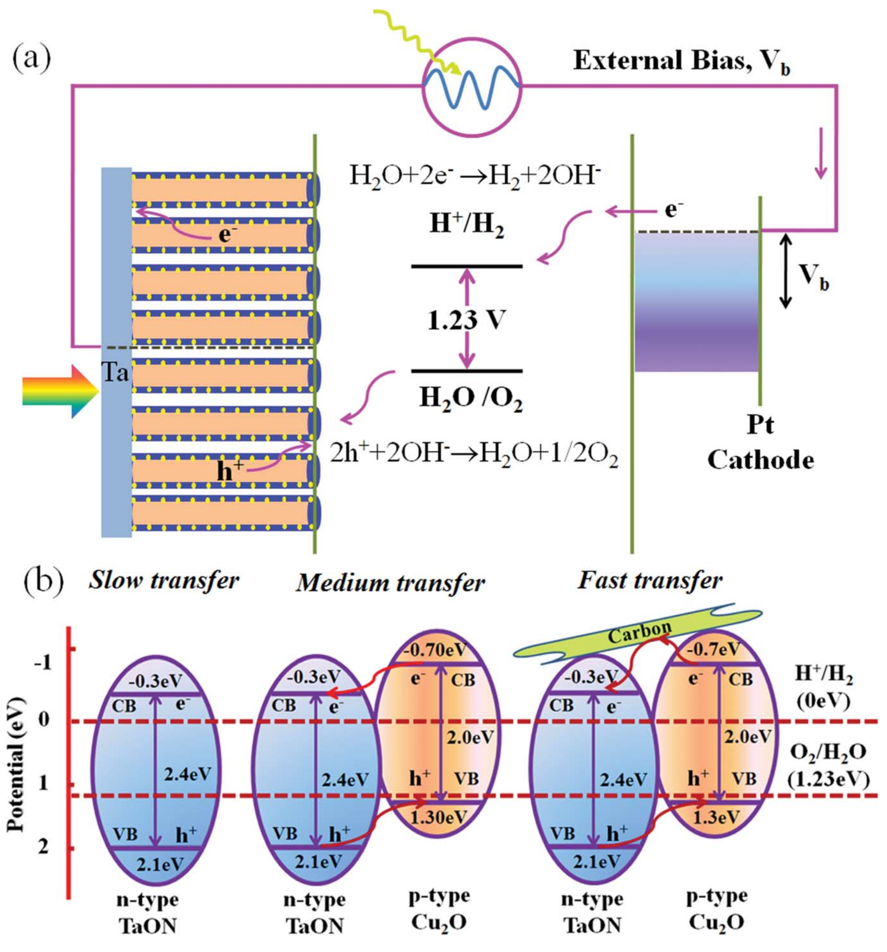


Fig. 7 Nyquist plot measured at an applied potential of 1.0 V *vs.* RHE under AM 1.5G simulated sunlight at 100 mW cm^{-2} for the TaON, Cu_2O/TaON and carbon–Cu_2O/TaON nanorod arrays. The inset is the magnified Nyquist plot for the various photoanodes.

electrodes to clearly show the important features of the entire plot on the same scale. When carbon and Cu_2O nanoparticles were introduced into the TaON system, it can be seen that another RC circuit occurs, indicating that the resistance decreased to a large extent. Compared to the TaON, Cu_2O/TaON and carbon–Cu_2O/TaON nanorod arrays under AM 1.5G simulated sunlight, it is obvious that the arches for the carbon–Cu_2O/TaON nanorod were much smaller than that of TaON, Cu_2O/TaON nanorod arrays, implying that the decoration with the carbon and Cu_2O nanoparticles significantly reduced the resistance to the movement of charge carriers due to the high electrical conductivity of carbon and the fast transport of charge carriers from the p–n junction formation of Cu_2O/TaON.

Based on the above-mentioned results, a possible PEC water splitting mechanism is proposed in Scheme 2. The TaON nanorod array with a relatively low IPCE was obtained due to the low transfer of charge carriers and the poor photostability of TaON itself. After decoration with Cu_2O nanoparticles, the resulting Cu_2O/TaON nanorod array was formed as a p–n heterojunction photoelectrode, *via* the chemical bath deposition process and subsequent annealing, with its enhanced photocurrent density resulting from the Cu_2O/TaON heterojunction structure more efficiently utilizing incoming light than when only TaON is used as the photoanode. Moreover, electrons from Cu_2O move toward the Ta substrate through TaON, and holes from TaON migrate to the surface of Cu_2O according to the



Scheme 2 Schematic for (a) the proposed mechanism of PEC water splitting and (b) the energy band structure of the TaON, Cu_2O/TaON and carbon–Cu_2O/TaON heterojunction photoanodes.

potential difference of Cu_2O ($E_g = 2.0$ eV, $E_{CB} = -0.7$ and $E_{VB} = +1.3$ eV)[37] and TaON ($E_g = 2.4$ eV, $E_{CB} = -0.3$ and $E_{VB} = +2.1$ eV).[14] Thus, the Cu_2O/TaON heterojunction photoanode not only absorbs more visible light, but also improves the efficiency of the charge separation across the junction structure and hence also improves the PEC water splitting performance. Besides, the relatively moderate rate of the charge transfer (noted as medium transfer) is ascribed to the enhanced photostability of TaON and the improved conductive interlayer between the photoelectrode and the substrate. The further improvement of the photocurrent and photostability by the ultrathin carbon layer in the carbon–Cu_2O/TaON photoanode is due to this carbon layer behaving as an electron conductor to enhance the electron–hole separation and easily transferring electrons from the conduction band of Cu_2O nanoparticles to the conduction band of the TaON nanorod array; these electrons then flow to the Ta substrate along the TaON nanorods, which provide a conductive electron transport "highway". The electrons ultimately transfer to the Pt counter electrode to reduce water. At the same time, the O_2 was produced by the oxidation of water using the accumulated holes on the surface of the Cu_2O nanoparticles protected by the encapsulating ultrathin graphitic carbon sheath. The nature of the carbon present in the carbon–Cu_2O/TaON samples with two peaks around 1341 and 1596 cm^{-1} was determined by Raman spectra (Fig. S11†), which confirmed the presence of sp^2 carbon-type structures within the carbonaceous wall of the carbon–Cu_2O/TaON samples. Thus, it is crucial to effectively separate the photogenerated electron–hole pairs for the enhancement of PEC activity. To a certain extent, the introduction of carbon and the formation of the Cu_2O/TaON heterojunction photoanode facilitate the enhancement of the efficiency of PEC performance. Therefore, the carbon–Cu_2O/TaON photoanode with the unique 1D heterojunction structure feature of being passivated with an ultrathin carbon sheath as a surface protection layer against corrosion—and integrating the advantages of enhanced light absorption, rapid charge separation and transport, large contact area and abundant hetero-interfaces—is more favorable to the migration of photogenerated electron–hole pairs, which contribute to the high PEC activity and photostability for water splitting.

In summary, we have demonstrated a promising strategy for fabricating a 1D carbon–Cu_2O/TaON photoanode. The resulting heterojunction exhibited significantly enhanced PEC water oxidation performance, including a large photocurrent density and high stability, which is superior to the reported TaON-based photoelectrodes. The enhanced light harvesting, efficient separation of photogenerated electron–hole pairs, fast charge transfer at the interface, and the unique 1D nanostructural heterojunction feature of the Cu_2O/TaON photoanode—along with its p–n heterojunction device being protected from electrolyte by being encapsulated in an ultrathin graphitic carbon sheath—all contribute to the excellent PEC activity for water oxidation. Our results have demonstrated remarkable strategies for designing and constructing heterojunction photoelectrodes for practical sunlight-driven PEC water splitting.

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